Application of Remote Sensing Technology in Monitoring Oil Pollution from Ship Grounding Incidents

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Marine oil pollution is a critical environmental issue, particularly following shipwrecks, where the inability to promptly seal all fuel tanks turns the sunken vessel into a latent ecological hazard. In such events, remote sensing technology provides an efficient and sustainable monitoring approach. Among these, satellite imagery is one of the most commonly used tools, with optical and radar sensors each possessing distinct characteristics and advantages. Additionally, drones enhance situational awareness by providing high-resolution data. This study examines the oil spill incident of the *Angel* vessel, which sank off the Port of Kaohsiung (Taiwan’s largest port and the biggest city in southern Taiwan) in July 2023, using various remote sensing technologies, including Sentinel-1, Sentinel-2, Landsat-8/9, and drone imagery, to analyze their applications and complementarities.

The *Angel* was a 172.17-meter-long, 25.41-meter-wide container ship that anchored approximately 2.8 nautical miles south of the Port of Kaohsiung’s southern breakwater on July 20, 2023. Due to suspected water ingress, the vessel sank at 00:19 on July 21, leaving an estimated residual fuel of 491.848 metric tons of heavy oils. Between July 21 and September 22, this study collected 29 optical satellite images, 12 radar satellite images, and drone footage covering 22 days to conduct cross-verification.

Analysis of the optical images revealed that 7 images were partially affected by cloud cover, making it difficult to delineate the oil spill, while 12 images were completely obscured, likely due to Taiwan’s terrain and climatic conditions. Among the identifiable images, both Sentinel-2 and Landsat-8/9 detected oil slicks, appearing silver in optical imagery as patches or streaks extending from the wreck site. For instance, on July 24, Sentinel-2 imagery showed the oil spill spreading southeast, covering an area of 20.3 km². However, drone footage indicated that dark oil patches were not visible in satellite images, possibly due to the oil’s volume and surface distribution characteristics.

In contrast, radar satellites, unaffected by cloud cover, provided distinct contrast between the oil spill and the surrounding sea surface. The Sentinel-1 image from July 27 clearly showed the oil spill spreading southeast, reaching up to 26.5 km from the wreck site and covering 10.7 km².

Meanwhile, drone imagery, offering high-resolution data, captured sheens extending southward from the wreck site on July 21, with dark oil appearing on July 22, exhibiting feathering patterns along the edges due to wind influence. Notably, drones also provided precise location tracking for floating cargo containers.

This study demonstrates the effectiveness of multi-source remote sensing technologies in monitoring oil spills from sunken vessels. Optical imagery offers intuitive spatial coverage but is limited in cloudy conditions. Radar imagery, while unaffected by clouds, is influenced by ocean surface roughness, requiring cautious interpretation. Drones provide high-resolution imagery but are constrained by operational range. The integration of these technologies enables comprehensive monitoring of oil spill dispersion, enhancing environmental surveillance. Future studies may explore the application of machine learning technique to improve automatic oil spill detection, further strengthening the role of remote sensing in marine pollution monitoring.

Keywords: Remote Sensing, Marine Oil Pollution, Oil Spill Monitoring

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